

Exam in "Economics & Law"

Winter Semester 2021/2022

Please read carefully the following instructions:

- This is a written exam based on 4 questions.
 - Each answer awards a different amount of points, as stated in brackets at the end of the question. The total points sum up to 100.
- This is an open-book exam. You may bring with you any written/printed material but you are not allowed to use a computer.

Questions:

Q1. "GreenTech" is a firm that produces solar panels. GreenTech has been enjoying high profits for the past year and therefore decided to donate money to the election campaign of the minister of finance. However, two changes have occurred in the market: First, a breakthrough in the production of silicon, a key material used in the production of solar panels, has made it cheaper and more readily available. Second, a new government policy has been implemented, which required the installation of solar panels in every governmental office.

(a) How should these events affect the equilibrium price and equilibrium quantity? **[10 points]**

First event: decrease in the price of an input = lower production costs = more supply. This decreases price but increases quantity.

Second event: governmental policy causes more demand. This increases price and quantity.

Hence, quantity increases but the effect on the price is ambiguous.

(b) Which economic mechanisms could explain the governmental policy mentioned in the question? **[10 points]**

- (1) Market failure (Externality): governmental agencies not caring enough about the environment, choosing not to install solar panels.
- (2) Public choice – the firm donated to the minister (rent seeking).

Q2. A young lawyer and an old lawyer are sitting on a bench. A potential client passes by. If both lawyers run toward the client at the same time, they slip and fall down, so that each of the lawyers gets a negative utility of -10. If only one lawyer approaches the client then he signs the client (whereas the other remains on the bench and gets zero). The old lawyer values signing the client at 20. The young lawyer values signing the client at 10. If no one approaches the client, both lawyers get a utility of zero.

(a) What is the Nash Equilibrium (or Nash Equilibria) of this game? **[10 points]**

Old lawyer/ young lawyer	Approach	Don't
Approach	-10, -10	20, 0
Don't	0, 10	0, 0

There are two Nash Equilibria in pure strategies: (Don't, approach) and (approach, Don't). Note: this is similar to an asymmetric chicken game, which we didn't cover in the course. There is also no need to check for mixed-strategy equilibrium (as it was also not covered in the course).

(b) Is the equilibrium (or equilibria) Pareto Efficient? Is it Kaldor-Hicks efficient? [5 points]

Both equilibria are Pareto-efficient. Only (approach, Don't) is KH efficient (yielding 20 in total).

(c) Suppose that the young lawyer is faster than the old lawyer, and can therefore act first. What is then the equilibrium? [5 points]

In a dynamic game, the young lawyer approaches, as this induces the old lawyer not to approach (this is similar to the market entry game we discussed in class). There is therefore only one subgame perfect equilibrium.

(d) Under the same assumption as in sub-question (c), suppose the old lawyer tells the young lawyer: "don't you dare approach the client! If you do, I will do so as well and we both slip and fall". Should the young lawyer worry about this threat? Explain your answer. [5 points]

If the young lawyer acts first, there is only one subgame perfect Nash equilibrium: the young approaches and the old does not. The threat of the old lawyer is, therefore, not credible, as it is not part of a sub-game perfect Nash equilibrium.

(e) If the two lawyers could bargain at zero transaction costs, what would be the outcome of the game? [5 points]

With zero TC, the Coase Theorem predicts the two layers will negotiate and opt for the KH solution. They will then split the surplus between them.

Q3. Jimmy is a judge in a criminal case. He decides to ask an AI-based chatbot what he recommends to do. The chatbot's output is the following:

"The defendant has long fingernails. It is well-known, that: 90% of people with long fingernails commit crimes; 10% of people with short fingernails do not commit crimes; and 0.1% of people have long fingernails. Furthermore, please note that 65% of judges using our services convict in these cases, will you do the same? By the way, there are two news articles about people with long fingernails who committed a crime (both of them robbed a bank). Also, you are our 25th user today. Thanks for using this Chatbot!"

Jimmy decides to convict the defendant and sentences him to 24 years in prison. The defendant wants to appeal and turns to you. Which arguments from behavioral economics could you use in your appeal (note: there is no need to use numerical calculations in your answer)? [25 points]

- (1) Representativeness heuristic – the judge neglected the base probability of 0.1% and focused only on the 90% (similar to the Guthrie et al. study about the barrels case).
- (2) Anchoring - the judge anchored on the irrelevant number 25 for sentencing (like the Englich et al. experiment on sentencing).
- (3) Availability heuristic (the news articles)
- (4) The AI is nudging the judge to convict (this is similar to the example on taxes seen in class).

[Grading was as follows: identifying correctly one of the issues – 7 points, only two of the issues – 13 points. Identifying three of the issues - 22 points. Identifying all four issues – 25 points]

Q4. In the developing country of “DevState”, there are large demonstrations protesting the inequality between employers and employees. The president of DevState turns to his advisors for advice:

Advisor 1: We should improve the working conditions for our workers by limiting their working week to only four days.

Advisor 2: I agree that we should help workers, but we need to intervene earlier – in the hiring stage. Currently older workers cannot get a job, so we should prohibit employers from asking about age.

Advisor 3: I think we should set up a platform for job-seekers. We can find a private firm to construct the platform and then regulate the prices it charges.

Advisor 4: I have a different idea. Why don't we demand our banks to give loans to poor entrepreneurs, so they can just build their own business?

Discuss (briefly!) whether these four advices make sense from an economic perspective. **[25 points]**

Advice 1 benefits all workers, but may discourage hiring [one can mention here that the effect may be similar to minimum wage + that production may decrease]. Advice 2 may help avoid discrimination, but may backfire (like ban-the-box): everyone who, e.g., looks old might not be hired. Advice 3 raises the tradeoff of infrastructure: the firm might not agree to invest if it knows its prices are later regulated. Advice 4 is about directed lending, which may be inefficient if the government does not have specialization to determine who should be funded+ there may be cannibalization of financial markets.